

The New Reproductive-Longevity Complex

Biotech Reproductive Technology, Germline Gene Editing, and Billionaire-Funded Longevity Science — State of the Field, August 2026

Prepared 2026-08-15. Method: synthesis of primary literature (Nature, NEJM, Nature Communications, Nature Medicine, Human Reproduction, Cell), clinical-trial registries, company press releases cross-checked against independent journalism (MIT Technology Review, STAT, Bloomberg, Science, TechCrunch, Fierce Biotech), and funding databases. Claims that rest only on company announcements or could not be independently verified are explicitly flagged as such throughout. Where credible sources conflict, the range is presented.

1. Executive Summary

Three previously separate fields — assisted reproduction, germline genome editing, and aging biology — are converging into a single capital ecosystem, funded to a striking degree by the same small group of technology billionaires, and advancing faster scientifically than the regulatory and ethical frameworks meant to govern them.

Key breakthroughs (2025–mid-2026):

- **Mitochondrial replacement worked in humans.** Newcastle University reported **8 healthy babies** born via pronuclear transfer to women carrying pathogenic mtDNA variants, published in *NEJM* (July 2025) — the first robust peer-reviewed clinical outcome data for any heritable germline intervention ([NEJM 2025](#)).
- **In-vitro gametogenesis (IVG) reached human proof-of-concept — with a hard failure mode.** OHSU’s Mitalipov lab generated 82 functional human oocytes from skin-cell nuclei via “mitomeiosis” (*Nature Communications*, Sept 2025), but **zero were chromosomally normal**; clinical use remains an estimated ~10 years away ([Nat Commun 2025](#)).
- **Base editing entered the human embryo.** Two independent June 2026 studies — Cambridge/Niakan targeting *NANOG* in *Nature*, and a Columbia/Egli-led consortium editing *PCSK9* and *HBG1/2* (bioRxiv preprint) — showed base editing avoids the chromosomal havoc of nuclease CRISPR, while **mosaicism and off-target effects remain unsolved** ([Nature 2026](#); [bioRxiv 2026](#)).
- **Partial epigenetic reprogramming reached first-in-human dosing.** Life Biosciences dosed the first patient with ER-100, an OSK gene therapy for optic neuropathies, in June 2026 (NCT07290244) — the first clinical test of the thesis behind roughly \$5B of “reprogramming” capital ([Life Biosciences 2026](#)).
- **AI-discovered longevity-adjacent drugs produced real clinical data.** Insilico Medicine’s rentosertib posted positive Phase 2a lung-function results in *Nature Medicine* (June 2025) and entered Phase 3; Insilico listed on the Hong Kong exchange in December 2025.

Capital flows. Longevity-biotech venture funding in 2025 is estimated between **\$5.7B (~170 deals)** and **\$8.5B (~325 deals)** depending on category definitions, with Q1 2026 up ~56% year-over-year. Marquee rounds include NewLimit’s **\$435M Series C at a \$3.1B valuation** (June 2026, Founders Fund-led, with Eli Lilly Ventures), Retro Biosciences’ reported ~\$1B raise at a \$1.8B pre-money valuation, and Gameto’s \$44M Series C (~\$127M total). A distinct, smaller wave of capital — Sam Altman and Brian Armstrong backing embryo-editing startup Preventive (~\$30M), Thiel-linked money in Nucleus Genomics — is now openly funding heritable-genetics ventures.

The central ethical tension. The most clinically mature technologies (Fertilo egg maturation, MRT, automated IVF) are targeted at disease and infertility and enjoy broad support. But the same technical stack — iPSC-derived gametes, base editing, whole-genome embryo screening — directly enables trait selection and heritable enhancement, and companies explicitly pursuing “superbabies” (Bootstrap Bio), IQ-ranked embryos (Nucleus, Herasight), and offshore germline editing (Preventive’s UAE exploration) emerged into the open in 2025–2026. Professional bodies (ASRM, ESHG, ISSCR) uniformly say these applications are not ready or not permissible; the commercial market is proceeding anyway, largely in regulatory gaps.

2. Reproductive Biotech Advances

2.1 In-vitro gametogenesis (IVG)

IVG — creating eggs or sperm from non-reproductive cells — is the field’s most consequential long bet: it would end gamete scarcity, extend fertility indefinitely, and enable same-sex genetic parenthood. Mouse IVG is essentially solved (Hayashi and Saitou produced fertile mice from stem-cell-derived eggs, including eggs from male mice in 2023). **Human IVG is not: as of August 2026, no laboratory has produced a mature, chromosomally normal (euploid), fertilization-competent human egg or sperm by any route, and no human has been born from a lab-made gamete.**

OHSU / Mitalipov — “mitomeiosis” (peer-reviewed). The landmark 2025 result came from Shoukhrat Mitalipov’s lab at Oregon Health & Science University (*Nature Communications*, 30 Sept 2025). Rather than differentiating stem cells into eggs, the team transplanted a somatic (skin-cell) nucleus into an enucleated donor oocyte and induced a synthetic reduction division — “mitomeiosis” — to halve the chromosome set. They generated **82 functional oocytes**; after fertilization, **~9% reached blastocyst** by day 6. The decisive negative: chromosome segregation was essentially random. **No analyzed embryo carried a correct euploid chromosome complement**, and the authors attribute the low blastocyst rate to aneuploidy plus incomplete epigenetic reprogramming (*Nat Commun* 2025). Mitalipov and independent commentators framed clinical application as **roughly a decade away**, contingent on solving chromosome pairing/segregation — STAT’s summary of expert consensus: it “partially works, and partially doesn’t.”

Conception Biosciences (company claim — not peer-reviewed). The Sam Altman- and Jaan Tallinn-backed startup (founded 2018, Matt Krisiloff) announced in **June 2026** that it had produced the **first early human egg cells (primary oocytes) from stem cells**, via blood cells → iPSCs → ovaroid (“mini-ovary”) culture. Two flags: these are **immature primary oocytes, not mature MII eggs**, and **no data have been published or peer-reviewed** — experimental detail is scarce. Widely cited funding is a ~\$20M round; larger totals (“\$50M+”) circulate but could not be verified against a primary source.

Male gametes — Paterna Biosciences (company claim — not peer-reviewed). Paterna announced it grew **functional human sperm entirely in vitro** and used them to fertilize eggs, forming embryos; it has raised **\$11.4M** (investors include IVIRMA/KKR and Mayo Clinic-affiliated funds) and is raising a \$12M Series A. The claim is unreplicated and unpublished — the earliest-stage headline in the field and the one most in need of independent verification.

Other players. **Dioseve** (Japan; transcription-factor-driven “DIOL” oocyte induction) raised >¥1B in a June 2026 Series A extension. **Ovelle Bio** claims the first meiotic oocyte-like cells from human iPSCs (George Church advising; unpublished). **Ivy Natal** and **Vitra Labs** (~\$8M raised) remain early preclinical. Academically, **Katsuhiko Hayashi** (Osaka) estimates **~7 years** to viable lab-grown human gametes; **Mitinori Saitou** (Kyoto/ASHBi) has reached human oogonia — but not mature oocytes — via PGC-like cell culture. The Nuffield Council on Bioethics published a dedicated IVG ethics review in July 2025.

Bottom line: IVG's unsolved core problem is a mature, euploid, epigenetically normal human gamete. Credible timelines to first clinical use cluster at **7–10+ years**, and that assumes a breakthrough on meiotic fidelity that has not yet occurred.

2.2 iPSC-based oocyte maturation — Gameto's Fertilo (most clinically advanced)

Gameto's **Fertilo** does not create eggs; it uses **iPSC-derived ovarian support cells** co-cultured with retrieved immature oocytes to mature them outside the body, cutting hormone stimulation by ~80% (2–3 days versus 10–14). It is the bridge product between conventional IVF and true IVG:

- **Peer-reviewed foundation:** a *Human Reproduction* study (67 participants) showed ~1.5× improved maturation versus commercial IVM media and ~2.5× improvement in euploid embryo formation.
- **First live birth:** Lima, Peru, **December 2024** — the world's first baby from an iPSC-derived-cell-assisted procedure.
- **US Phase 3:** FDA cleared the IND on **30 January 2025** — the first iPSC-based therapy to reach Phase 3 in the US — across ~15 sites, primary endpoint clinical pregnancy at 12 weeks; first patients recruited mid-2025.
- **Funding and rollout:** **\$44M Series C** (Aug 2025, Overwater Ventures), ~**\$127M** total. Company-reported (not independently audited): **5 babies born, 20+ pregnancies**, with commercial use or clearance in Peru, Mexico, Australia, Japan, India, Singapore and several Latin American markets.

Flag: the birth/pregnancy tallies and country clearances are press-release figures; pivotal Phase 3 efficacy data are not yet published.

2.3 Full lab automation and AI in IVF

- **Conceivable Life Sciences (AURA).** The first baby conceived via a **fully automated, remotely operated ICSI procedure** was born **10 April 2025** (case report in *Reproductive BioMedicine Online*; team led by embryologist Jacques Cohen). AURA automates 200+ manual lab steps. Company-reported pilot outcomes: 100+ patients, 1,000+ eggs, 60+ pregnancies, 30+ live births. Funding: **\$18M Series A (Jan 2025, ARTIS Ventures)**, >\$39M total, strategic backing from IVI RMA; Mexico City pilot with US launch targeted 2026. Co-founder Martin Varsavsky ties it into the Prelude fertility-clinic network.
- **Overture Life.** **\$57M total** raised (April 2025 round with GV and Khosla), plus a **€20M European Investment Bank loan** (Nov 2025); US CLIA license for its non-invasive metabolomic embryo test; DaVitri vitrification robots deployed; first commercial automated-IVF agreement with Memorial Hospital Istanbul (July 2025).
- **Fertilis** (Australia) published microfluidic automation improving oocyte recovery in *Nature Medicine* (2026) — notable as automation evidence in a top-tier journal rather than a press release.
- **AI embryo selection (AIVF, Embryonics, Genea and peers)** remains clinically contested: most validation is retrospective, and there is still **no strong RCT evidence that AI selection improves live-birth rates** over conventional morphology. Treat vendor performance claims accordingly.

The strategic significance of automation is less per-cycle novelty than cost structure: robotic labs plus abbreviated stimulation (Fertilo-style) could cut IVF's cost several-fold, which is the precondition for the "IVF at scale" future that both fertility investors and embryo-screening companies are betting on.

2.4 Mitochondrial replacement therapy (MRT)

The UK's Newcastle programme delivered the field's most rigorous clinical result: two *NEJM* papers (17 July 2025) reporting **22 women** treated by pronuclear transfer and **8 live births** (including twins). Infant blood mutant-mtDNA loads ranged from **undetectable to 16%** — a 77–100% reduction versus the maternal variant. All eight children were healthy at birth with normal development; two showed elevated heteroplasmy (12% and 16%), which the authors judge not to be donor-carryover but which keeps the “reversal” risk under active surveillance. Two children had treatment-responsive or self-resolving health issues not attributed to MRT (*NEJM* 2025; [Science Media Centre expert reaction](#)).

Australia is following via Maeve's Law: the Monash-led **mitoHOPE** programme (A\$15M) received its first licence in Nov 2025 and performed Australia's first maternal spindle transfer with human cells in **March 2026** — still preclinical, no births yet. **In the US**, MRT remains effectively banned by the annually renewed FDA appropriations rider barring review of heritable-modification applications (see §3.4; current-year rider status flagged as not independently confirmed).

2.5 Adjacent: reproductive longevity therapeutics

- **Columbia VIBRANT trial**: a randomized, placebo-controlled pilot of weekly low-dose **rapamycin in 50 women aged 38–45** (Suh/Williams). Interim institutional reporting suggests **~20% slowing of ovarian follicle loss**. *Flag: pilot-stage, interim, not yet a completed peer-reviewed RCT — but it is the first credible pharmacological attempt to slow ovarian aging in humans.*
- **Oviva Therapeutics** (anti-AMH pathway, OVI-586, preclinical) was **acquired by Granata Bio in April 2025** — evidence the “delay menopause” thesis is consolidating into the fertility industry rather than standing alone.
- **Ovo Labs** (ex vivo egg-rejuvenation biologics) and **Celmatix** (AMHR2 agonists) remain preclinical.
- **Uterine transplantation** has quietly become established medicine: ~80 procedures and ~40 live births globally, with pooled live-birth-per-embryo-transfer around 22% and Australia's first birth reported in 2025.

A claimed Gameto ovarian-longevity program named “Ameca” could not be verified and should be treated as unconfirmed.

3. Germline Gene Editing

3.1 The June 2026 base-editing embryo studies

Two independent studies, published within weeks of each other, moved human embryo editing from nuclease CRISPR (which causes double-strand breaks and chromosomal chaos) to **base editing** (single-nucleotide chemistry, no double-strand break):

Cambridge — NANOG (Niakan lab, *Nature*, June 2026). The first use of base editing in human embryos for developmental research. Silencing *NANOG* prevented epiblast formation — the lineage from which all body tissues derive — and revealed that, unlike in mice, human embryos have **no compensatory backup** for *NANOG* loss. The work was explicitly a research knockout tool, not a reproductive step; base editing was chosen precisely because nuclease CRISPR's chromosomal damage made it unsuitable (*Nature* 2026).

Columbia/Egli consortium — PCSK9 and HBG1/2 (bioRxiv preprint, June 2026, not yet peer-reviewed). An industry-academic collaboration used adenine base editors to make disease-relevant edits (cholesterol and fetal-hemoglobin targets), in some cases editing two genes in the same embryo. Headline positive: efficient editing with **no**

aneuploidy, chromosomal abnormalities, or large deletions — some embryos were uniformly edited. Headline negatives: **substantial mosaicism persisted** even with single-cell-stage editing (a just-fertilized egg already carries up to four genome copies; the editor frequently missed one), plus measurable off-target and unintended on-target edits. The preprint suggests editing would need to occur within ~5–12 hours of fertilization to suppress mosaicism. Egli’s own framing — “this will change the discussion on heritable editing” — was paired with his public call for “tough conversations on ethical boundaries” ([bioRxiv 2026](#); [STAT 2026-06-24](#)). Reporting indicates **Nucleus Genomics** is a collaborator on this research line (reported, not fully confirmed).

Data caveat: exact efficiency, mosaicism, and off-target percentages from both papers could not be extracted from paywalled/blocked primary texts for this report; qualitative conclusions are well-corroborated across Science, C&EN, CNN and STAT coverage, but precise figures here should be re-checked against the papers directly.

Interpretation. Base editing has removed the most catastrophic technical objection to embryo editing (chromosomal destruction) while leaving the two objections that matter clinically — **mosaicism** (an embryo whose cells carry different genomes defeats both safety validation and the intended benefit) and **off-target risk** — unresolved. Both teams state clearly that reproductive use is not justified. That did not stop the studies from being read, accurately, as shrinking the technical distance to heritable editing.

3.2 Companies pursuing heritable editing

- **Preventive** (San Francisco, founded ~2025). CEO **Lucas Harrington** (Mammoth Biosciences co-founder, Doudna lab). **~\$30M raised; investors include Sam Altman and Brian Armstrong**. Public-benefit corporation; stated mission is rigorous *preclinical* determination of whether preventive germline editing can be made safe, with a pledge not to proceed clinically otherwise — while **exploring clinical work in the UAE**, where embryo editing is not prohibited ([MIT Technology Review, 2025-10-31](#)).
- **Bootstrap Bio** (California, seed-stage). Openly interested in **intelligence enhancement**; investor materials floated edited-embryo pregnancies in **2026–2027 in Honduras** (including the Próspera special economic zone) ([Bloomberg, 2025-06-25](#)). This is the field’s clearest enhancement-first venture and its most acute regulatory-arbitrage case.
- **Manhattan Genomics** → **Origin Genomics** (Cathy Tie). Manhattan Genomics (2025, with ex-Colossal scientist Eriona Hysolli, and publicly entangled with **He Jiankui**) **ceased operations by December 2025**. Tie relaunched as **Origin Genomics** (March 2026): US-only, IRB-overseen “responsible germline gene correction for severe inherited disease,” plus a stated intent to offer MRT domestically if Congress permits it. (The “Heliospect” name sometimes attached to Tie is a conflation — Heliospect Genomics is a separate UK-linked polygenic-screening operation.)
- **He Jiankui himself** remains in limbo: claims of an Austin, Texas lab and a \$1M US backer are his own and unconfirmed; SCMP reporting indicates he is barred from leaving China. Treat all “He restarts” stories as aspirational until independently verified.

3.3 Polygenic embryo screening (PGT-P): the commercial front line

While germline editing stays preclinical, embryo *selection* is a live consumer market:

- **Orchid Health** (Noor Siddiqui, founded 2019; ~\$16.5M raised). Whole-genome sequencing of IVF embryos at **~\$2,500/embryo**; first baby born April 2024; high-profile Silicon Valley clientele. In 2025 the **Psychiatric Genomics Consortium publicly stated Orchid’s psychiatric risk scores use consortium data in violation of use restrictions** that bar embryo-screening applications — the sharpest data-governance dispute in the sector ([Science](#)).

- **Nucleus Genomics** (Kian Sadeghi; ~\$32M raised, backers include Founders Fund and Seven Seven Six). Launched “**Nucleus Embryo**” in June 2025 at \$5,999, ranking embryos not only on disease risk but on **IQ, height, BMI, longevity, and cosmetic traits** — triggering the year’s fiercest “designer baby” backlash, a *Scientific American* rebuttal (“misleading, unethical — and not even possible”), and a trade-secrets lawsuit from Genomic Prediction (Aug 2025).
- **Herasight** (founded 2023, out of stealth 2025). ~\$50,000 per analysis including lifespan and IQ predictions; technically distinguished by *ImputePGTA* (reconstructing embryo genomes from parental WGS + routine PGT-A data; medRxiv preprint Nov 2025). Backers undisclosed.
- **Genomic Prediction / LifeView** remains the longest-running PGT-P provider (Embryo Health Score licensed to clinics).

The science says the product oversells. The quantitative literature — Karavani et al. (*Cell* 2019: top-embryo selection yields ~2.5 cm height or ~2.5 IQ points expected gain), Turley et al. (*NEJM* 2021: within-family predictive power shrinks, plus pleiotropy and ancestry-portability problems) — implies modest, uncertain gains from realistic embryo cohorts. Industry cites a genuine counterpoint (Moore et al.’s sibling-pair validations showing minimal within-family attenuation for most disease scores), so this is a live dispute — but for *behavioral/cognitive* traits, the mainstream statistical-genetics community (Shai Carmi, Sasha Gusev among the clearest voices) regards IQ-ranking of embryos as scientifically unsupported. **ASRM’s Ethics and Practice Committees concluded in December 2025 that PGT-P is not ready for clinical use; ESHG had already called it “unproven and unethical.”** The market is operating against explicit professional-society consensus.

3.4 Law and governance

- **US:** the FDA appropriations rider (since FY2016) bars the agency from even reviewing applications involving heritable modification — blocking both germline editing and MRT. It is renewed annually and has been contested in committee; **its precise FY2026 status could not be primary-source-verified for this report** (flagged), but no reporting indicates it has lapsed.
- **UK:** research editing under HFEA licence (14-day limit) is legal — the Niakan study’s pathway; implantation of edited embryos is criminal.
- **China:** post-He Jiankui criminal liability for illegal implantation.
- **The gap:** none of this binds the UAE, Honduras, or other jurisdictions Silicon Valley germline ventures are openly evaluating. **ISSCR** (heritable editing “currently not permitted”) and the **NASEM/Royal Society 2020 framework** remain the reference standards, with no clinical green light anywhere.

4. Billionaire Capital and Longevity Science

4.1 The majors

Altos Labs — \$3.0B (2022), the field’s flagship. Backers reported to include Jeff Bezos and Yuri Milner; leadership Hal Barron (CEO) and Rick Klausner (chief scientist). Still **entirely preclinical** with no disclosed IND. The 2025 signal was the appointment of **Joan Mannick as Chief Medical Officer** — read across the industry as the pivot from open-ended reprogramming research toward clinical development, with programs in ex vivo organ rejuvenation and in vivo partial reprogramming and possible first studies in the 2026 window ([Altos press release](#)). *Claims of a named 2026 “Sentinel Cell” program circulate only in low-quality aggregators — unverified.*

NewLimit — the fastest-scaling reprogramming company. Co-founded 2021 by **Brian Armstrong** (Coinbase) and Blake Byers, with Jacob Kimmel leading R&D. **\$130M Series B (May 2025, Kleiner Perkins)**, then a **\$435M Series C at a \$3.1B valuation (June 2026)** led by Founders Fund with Thrive, Greenoaks and **Eli Lilly Ventures** — a valuation triple in ~12 months and the clearest pharma endorsement of epigenetic reprogramming to date. Approach: LNP/mRNA delivery of transcription-factor cocktails to rejuvenate specific cell types, hepatocytes first. **Preclinical; first Phase 1 planned 2027 in MASH/fatty liver (STAT 2026-06-02).** (Note: frequently misattributed to Sam Altman — Altman’s vehicle is Retro.)

Retro Biosciences — Sam Altman’s \$180M bet, scaling to ~\$1B. Mission: +10 healthy years. TechCrunch reported a **\$1B Series A raise** beginning Jan 2025; by May 2026 an initial close at a **\$1.8B pre-money valuation** was reported — *whether the full \$1B closed is not cleanly verified*. Lead clinical asset **RTR242**, an oral autophagy-restoring small molecule for **Alzheimer’s**, entered Phase 1 around late 2025 (no NCT ID retrieved — flagged). Retro’s collaboration with OpenAI produced **GPT-4b micro**, a protein-design model whose engineered Yamanaka-factor variants (“RetroSOX,” “RetroKLF”) reportedly boosted reprogramming-marker expression >50× in vitro — announced via MIT Tech Review/OpenAI, **not peer-reviewed (MIT Tech Review 2025-01-17).**

Life Biosciences — the first-in-human milestone. Co-founded by David Sinclair; >\$400M raised historically. Its **ER-100** (AAV2-delivered OSK partial reprogramming) for optic neuropathies (NAION and glaucoma) received **IND clearance 15 Jan 2026** and **dosed its first Phase 1 patient 9 June 2026 (NCT07290244)** — the first human test of partial epigenetic reprogramming, and arguably the single most important clinical event in longevity biotech to date. (Earlier claims of “early 2025 dosing” reflect the original plan, not what happened.)

Insilico Medicine — the AI-pharma proof point. **Rentosertib** (TNIK inhibitor, AI-discovered target *and* AI-designed molecule) posted positive Phase 2a results in IPF — +98.4 mL FVC versus −20.3 mL placebo over 12 weeks, n=71 — published in *Nature Medicine*, **June 2025**; Phase 3 initiated. **IPO on the Hong Kong Main Board 30 Dec 2025 (03696.HK)**, raising ~\$293M — the year’s largest HK biotech listing.

Calico — the cautionary tale. After ~11 years and up to \$3.5B of committed AbbVie partnership economics, its eIF2B activator **fosigotifator failed the HEALEY ALS platform trial** (Jan 2025), and **AbbVie terminated the collaboration in November 2025** (~100 scientists laid off). Partial redemption: fosigotifator earned **FDA Breakthrough Therapy Designation for Vanishing White Matter disease** (June 2026). Calico’s opacity relative to its scale remains the standing criticism.

Chan Zuckerberg Initiative / Biohub. November 2025 pivot: CZI concentrated essentially all philanthropy into **AI-driven biology** (“cure, prevent or manage all disease”), doubling a ~\$4B decade commitment (~\$1B/year), absorbing EvolutionaryScale’s team and targeting 10,000 GPUs by 2028 for open **virtual-cell models**. Not a longevity funder by framing, but building infrastructure the aging field will run on.

4.2 The rest of the map

- **Hevolution Foundation** (Saudi): headline **\$1B/year** commitment; **actual disbursement ~\$400M cumulative** across 230+ grants, with 2025 commitments including a \$230M geroscience program — and acknowledged internal restructuring that may slow disbursement. The pledge-to-cash gap is the key fact.
- **XPRIZE Healthspan**: \$101M, 7-year prize to restore ≥10 years of muscle/cognition/immune function; 2025 milestone awards paid, finalist teams (sources vary between 10 and 20 across rounds — likely semifinal vs final cohorts) begin trials with data due April 2026; UCSD named testing hub.
- **Peter Thiel**: long-standing Methuselah Foundation/SENS (now Aubrey de Grey’s **LEV Foundation**) donor and cryonics enthusiast; portfolio company **Unity Biotechnology was delisted from Nasdaq (Aug 2025) and moved**

to **liquidation** — the decade’s clearest senolytics failure.

- **Immortal Dragons** (Singapore, Boyang Wang, ~\$40M): explicitly “make death optional,” impact-over-returns, 15+ portfolio companies including organ-fabrication startup Frontier Bio. Small capital, outsized rhetoric — the purest example of radical-longevity branding.
- **BioAge Labs** (NASDAQ: BIOA): lead obesity asset azelaprag **terminated Jan 2025** after liver toxicity; pivoted to NLRP3 inhibition with a Novartis deal — a reminder that public-market longevity plays get judged as ordinary biotechs.
- **Rubedo Life Sciences**: \$40M Series A; senolytic **RLS-1496** cleared IND (Sept 2025) with **first human data April 2026** in actinic keratosis — keeping the senolytic hypothesis alive at small scale after Unity.
- **Bryan Johnson**: told Wired (July 2025) he is “so close” to shutting or selling **Blueprint**, pivoting to the “**Don’t Die**” movement launched as a quasi-religion in March 2025. Culturally enormous, clinically nil.

4.3 Healthspan versus radical longevity — the honest ledger

Institutional capital (Altos, NewLimit, Hevolution, XPRIZE, CZI, every VC-backed clinical program) frames its work as **healthspan and disease**, because “aging” is not an FDA-approvable indication — hence trials in NAION, MASH, IPF, ALS, actinic keratosis. Radical life-extension language is concentrated in founder-driven or fringe vehicles (Immortal Dragons, Don’t Die, LEV Foundation). The scientific debates that will decide who is right: whether **partial reprogramming translates** beyond mice (context-dependence, tumorigenicity of OSK — Life Bio’s Phase 1 is the first real answer), and whether **epigenetic clocks** — population-valid but individually unreliable — can ever become regulator-accepted endpoints. Until an aging surrogate endpoint exists, every “longevity” company is legally a disease company.

5. Leading Companies Table

Company	Focus	Total raised (approx.)	Key backers	Stage (Aug 2026)
Gameto	iPSC ovarian support cells (Fertilo) for egg maturation	~\$127M	Overwater, atai founders	Phase 3 (US) ; commercial ex-US; 5 births (co. figures)
Conceivable Life Sciences	Fully automated AI IVF lab (AURA)	>\$39M	ARTIS, IVI RMA	Clinical pilot; first automated-ICSI baby Apr 2025
Overture Life	IVF automation + non-invasive embryo testing	\$57M + €20M EIB	GV, Khosla	Commercial (Istanbul, Dallas)
Conception Biosciences	Human eggs from stem cells (IVG)	~\$20M verified	Sam Altman, Jaan Tallinn	Preclinical; unpublished primary-oocyte claim (Jun 2026)
Paterna Biosciences	Lab-grown human sperm	\$11.4M	IVIRMA/KKR, Mayo	Preclinical; unpublished world-first claim
Dioseve	Transcription-factor oocyte induction	>\$7M + ¥1B ext.	Japanese VCs	Preclinical
Orchid Health	Whole-genome embryo screening	~\$16.5M	a16z angels et al.	Commercial (~\$2,500/embryo)
Nucleus Genomics	Consumer WGS + embryo trait ranking	~\$32M	Founders Fund, Seven Seven Six	Commercial (\$5,999); litigation

Company	Focus	Total raised (approx.)	Key backers	Stage (Aug 2026)
Herasight	Polygenic embryo prediction (incl. IQ)	Undisclosed	Undisclosed	Commercial (~\$50K)
Preventive	Heritable embryo editing (preclinical)	~\$30M	Sam Altman, Brian Armstrong	Preclinical; UAE exploration
Bootstrap Bio	Germline enhancement ("superbabies")	Seed	Undisclosed	Preclinical; Honduras plans (claimed)
Altos Labs	Partial reprogramming / cell rejuvenation	\$3.0B	Bezos, Milner (reported)	Preclinical; CMO hired 2025
NewLimit	Epigenetic reprogramming (liver first)	~\$700M (\$3.1B val.)	Armstrong/Byers; Founders Fund; Lilly	Preclinical; Ph1 planned 2027
Retro Biosciences	Reprogramming, plasma, autophagy	\$180M + ~\$1B raise (partial close verified)	Sam Altman	Phase 1 (RTR242, Alzheimer's)
Life Biosciences	OSK partial reprogramming (eyes)	>\$400M	Sinclair-founded, various	Phase 1 dosing Jun 2026 (NCT07290244)
Insilico Medicine	AI drug discovery (TNIK/IPF)	~\$400M + \$293M IPO	Warburg, HK listing	Phase 3 ; public (03696.HK)
Calico	Aging biology; eIF2B	Alphabet-funded (~\$3.5B AbbVie deal ended)	Alphabet	Ph1b/2 (VWM); AbbVie exit Nov 2025
Rubedo	Selective senolytics	\$40M+	Khosla, Hevolution	Phase 1b/2a; first human data Apr 2026
BioAge	NLRP3 / metabolic aging	Public (BIOA)	IPO 2024	Ph1; lead asset terminated 2025

Funding figures are best-verified approximations; private-company numbers derive from press releases and funding databases and may lag reality.

6. High-Signal Researchers, Founders, and Their X Accounts

Handles verified against x.com URLs or explicit citations as of Aug 2026; several prominent scientists (Mitalipov, Hayashi, Saitou, Amato) have **no verifiable X presence** — monitor their institutions instead.

Account	Who	Why high-signal
@EgliDieter	Dieter Egli, Columbia	Author of the 2026 embryo base-editing preprint; rigorous, low-volume posts on embryo genome stability
@NiakanLab	Kathy Niakan's lab, Cambridge	The lab behind the <i>NANOG</i> Nature paper; UK embryo-research governance
@clarklabucla1	Amander Clark's lab, UCLA	IVG and reproductive stem-cell policy
@ShaiCarmi	Shai Carmi, Hebrew U.	The key statistician on what embryo selection can and cannot deliver
@SashaGusevPosts	Sasha Gusev, Dana-Farber	Sharpest technical critic of IQ/polygenic embryo hype (Substack: The Infinitesimal)
@pknoepfler	Paul Knoepfler, UC Davis	The Niche blog; watchdog on stem-cell clinics, IVG, embryo editing

Account	Who	Why high-signal
@HankGreelyLSJU	Hank Greely, Stanford Law	Leading legal/ethics scholar on ART and germline law
@UrnovFyodor	Fyodor Urnov, IGI Berkeley	CRISPR-cures frontline; forceful on somatic-vs-germline boundaries
@geochurch	George Church, Harvard	Gene editing and synthetic biology; advises multiple reproductive/longevity startups
@davidasinclair	David Sinclair, Harvard	Reprogramming/Information Theory of Aging; huge reach — read alongside his critics
@mkaeberlein	Matt Kaeberlein, Optispan	The field's most rigorous skeptic (rapamycin, biomarkers)
@prof_horvath	Steve Horvath, Altos	Inventor of the epigenetic clock
@DrMorganLevine	Morgan Levine, Altos	PhenoAge; biological-age measurement
@jpsenescence	João Pedro de Magalhães	GenAge/AnAge databases; genomics of aging
@biogerontology	Alex Zhavoronkov, Insilico	AI-drug-discovery × longevity industry commentary
@sama	Sam Altman	Backer of Retro, Conception, Preventive — capital signals
@bryan_johnson	Bryan Johnson	Don't Die movement; the culture of longevity
@noor_siddiqui_	Noor Siddiqui, Orchid	Embryo-screening market's most visible founder
@KianSadeghi5	Kian Sadeghi, Nucleus	The IQ-screening controversy, firsthand
@DinaRadenkovic (+ @GametoGen)	Dina Radenkovic, Gameto	Fertilo clinical progress and ovarian aging
@martinvars	Martin Varsavsky	Fertility-industry economics; Conceivable/Prelude
@bettslacroix	Joe Betts-LaCroix, Retro	Reprogramming and healthspan-extension programs
@jacobkimmel	Jacob Kimmel, NewLimit	Posts actual reprogramming screen data
@LauraDeming	Laura Deming, Cradle	Longevity-VC pioneer; now reversible cryopreservation
@KarlPfleger	Karl Pfleger	agingbiotech.info — the best structured industry tracker
@antonioregalado	Antonio Regalado, MIT Tech Review	Broke the He Jiankui story; the essential embryo-editing/IVG beat reporter
@celinehaloua	Celine Haloua, Loyal	The pragmatic FDA path to an aging drug (dogs first)

Caveat: @EricVerdin (Buck Institute) was reportedly hacked and re-created in early 2026 — verify before relying on it. Follower counts and account states drift; re-verify handles before citing.

7. Strategic Analysis

7.1 Technical readiness versus hype

A sober readiness ladder as of August 2026, from proven to speculative:

1. **Proven and clinical:** MRT (8 births, NEJM); Fertilo egg maturation (Phase 3, peer-reviewed mechanism); automated ICSI (live births); uterine transplant.

2. **Credible and in trials:** Life Bio's OSK reprogramming (Phase 1 just dosed); Insilico's rentosertib (Phase 3 — AI-discovered, longevity-adjacent rather than anti-aging); rapamycin for ovarian aging (pilot); Rubedo senolytics (Phase 1b/2a).
3. **Preclinical with strong science:** NewLimit and Altos reprogramming; base editing in embryos (research only).
4. **Proof-of-concept with unsolved failure modes:** human IVG (zero euploid embryos in the best published attempt).
5. **Marketed beyond the evidence:** polygenic trait/IQ embryo ranking (against ASRM/ESHG consensus); AI embryo selection (no live-birth RCT superiority).
6. **Unpublished claims:** Conception's oocytes, Paterna's sperm, Ovelle's meiotic cells, Retro/OpenAI's protein-design results.

The pattern: **the closer a technology is to conventional medicine (a disease, a registrable endpoint), the more real it is.** The most-hyped categories are precisely those without registrable endpoints (aging itself) or without validated predictive power (behavioral PGS).

7.2 Convergence of reproductive longevity and organismal aging

The two fields are becoming one investment thesis. The ovary is the fastest-aging human organ, making ovarian aging both an unmet medical need and the most tractable *proving ground for geroscience* (rapamycin/VIBRANT; AMH modulation; Dioseve's framing as "reproductive longevity"). Conversely, IVG plus reprogramming implies something more radical: if gametes can be manufactured from somatic cells, reproductive age ceases to be biologically bounded — the same Yamanaka-factor toolkit underlies Altos's rejuvenation and Conception's eggs. The same investors (Altman across Retro/Conception/Preventive; Armstrong across NewLimit/Preventive) hold positions across both, which is why policy analysis should treat "reproductive longevity" and "anti-aging" as a single complex rather than adjacent sectors.

7.3 Regulatory and ethical gaps

- **The US governs by appropriations rider,** not statute — a one-line renewal battle each year decides MRT and germline-editing policy, with no affirmative framework for even the well-evidenced case (MRT post-NEJM).
- **Jurisdiction shopping is now explicit:** Fertilo commercialized in Peru/Mexico before US approval; Preventive evaluates the UAE; Bootstrap Bio names Honduras. Ethics frameworks (ISSCR, NASEM) bind no one.
- **Consumer genomics is effectively unregulated:** PGT-P products with IQ claims face no premarket review; the constraints operating today are data-use licenses (the PGC-Orchid dispute) and trade-secret suits — not health regulators.
- **The equity problem compounds:** at \$2,500–\$50,000 per screen atop \$20K+ IVF cycles, these technologies are currently instruments of the wealthy; if PGS worked as advertised (it mostly doesn't, for traits), it would convert existing wealth inequality into heritable biological advantage — the substantive core of the "designer baby" objection, distinct from the safety question.

7.4 Forecast 2026–2030 (base case)

- **2026–2027:** Fertilo Phase 3 readout — if positive, the first FDA-approved iPSC-derived-cell fertility product (~2027–28). Life Bio ER-100 Phase 1 safety data — the single most consequential readout for the reprogramming thesis. NewLimit IND/Phase 1 (2027). Retro RTR242 Phase 1 data. Expect at least one polygenic-screening company to face FTC or state-level action, and continued PGT-P professional-society pressure.
- **2027–2028:** First Australian MRT pregnancies (mitoHOPE). Automated-IVF platforms enter US clinics at scale; IVF cost curve begins bending. Base-editing embryo research consolidates mosaicism-timing findings; expect a

formal push (Egli and others have already called for it) to renegotiate the international moratorium framing. Rentosertib Phase 3 readout — a positive result would validate the AI-discovery stack pharma-wide.

- **2029–2030:** Realistic best case for human IVG is a **euploid, mature lab-derived human oocyte demonstrated academically** — not a birth; clinical IVG remains post-2032 even optimistically (Hayashi ~7 years from 2025; Mitalipov ~10). XPRIZE Healthspan final judging (2030) forces the field’s first head-to-head functional-aging evidence. The probability that someone attempts an edited-embryo pregnancy in a permissive jurisdiction by 2030 is uncomfortably material — Bootstrap Bio has named dates — and the field’s governance credibility will be defined by how (and whether) the mainstream reacts before it happens, not after.
 - **What would falsify the optimists:** ER-100 safety failure (tumorigenicity would chill reprogramming capital broadly); Fertilo Phase 3 miss; continued absence of any peer-reviewed data from Conception/Paterna by 2028.
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8. Annotated Source List

Peer-reviewed primary literature

1. **Mitalipov lab** — *Induction of experimental cell division to generate cells with reduced chromosome ploidy*, **Nature Communications** (2025-09-30). [nature.com/articles/s41467-025-63454-7](https://www.nature.com/articles/s41467-025-63454-7). *The definitive human-IVG proof-of-concept; also the definitive statement of its aneuploidy problem.*
2. **Newcastle MRT team** — pronuclear transfer outcomes, **NEJM** (2025-07-17). [NEJMoa2503658](https://www.nejm.org/doi/full/10.1056/NEJMoa2503658) and companion [NEJMoa2415539](https://www.nejm.org/doi/full/10.1056/NEJMoa2415539). *Eight births; heteroplasmy quantified; the clinical benchmark for germline interventions.*
3. **Niakan lab** — *Base editing reveals an essential role for NANOG in human embryogenesis*, **Nature** (2026-06). [nature.com/articles/s41586-026-10792-1](https://www.nature.com/articles/s41586-026-10792-1). *First base-edited human embryos; developmental biology application.*
4. **Egli et al.** — *Efficient base editing and development in human embryos without chromosomal alterations*, **bioRxiv** (2026-06, preprint — not yet peer-reviewed). [biorxiv.org](https://www.biorxiv.org/). *Disease-target embryo editing; mosaicism data; read with the Science and STAT coverage for context.*
5. **Gameto** — OSC co-culture oocyte maturation, **Human Reproduction**. Via gametogen.com/publications. *The peer-reviewed mechanism behind Fertilo.*
6. **Insilico** — rentosertib Phase 2a, **Nature Medicine** (2025-06). Via insilico.com/news. *AI-discovered drug clinical proof-of-concept.*
7. **Karavani et al.**, **Cell** (2019) [10.1016/j.cell.2019.10.033](https://doi.org/10.1016/j.cell.2019.10.033); **Turley et al.**, **NEJM** (2021) [NEJMs2105065](https://doi.org/10.1056/NEJMs2105065). *The quantitative case against trait-based embryo selection; still the anchor citations.*
8. **Fertilis** — microfluidic oocyte recovery, **Nature Medicine** (2026). [nature.com/articles/s41591-026-04207-x](https://www.nature.com/articles/s41591-026-04207-x).
9. **Uterine transplant systematic reviews** — [PubMed 39579025](https://pubmed.ncbi.nlm.nih.gov/39579025/); Brännström, *Reprod Med Biol* (2025).

Registries and official documents

1. **NCT07290244** — Life Biosciences ER-100 Phase 1. clinicaltrials.gov/study/NCT07290244.
2. **ASRM Ethics/Practice Committees on PGT-P** (2025-12) — [asrm.org press release](https://asrm.org/press-releases). *The clearest professional-society verdict.*
3. **ISSCR Guidelines** — [isscr.org/guidelines](https://www.isscr.org/guidelines); **NASEM Heritable Human Genome Editing (2020)** — [NBK565931](https://www.nasem.org/heritable-human-genome-editing).
4. **Nuffield Council on Bioethics, IVG report** (2025-07) — nuffieldbioethics.org PDF.

Journalism and investigations

1. **Antonio Regalado, MIT Technology Review** — Preventive exposé (2025-10-31) [technologyreview.com](https://www.technologyreview.com); OpenAI/Retro GPT-4b micro (2025-01-17). *The indispensable beat*.
2. **Bloomberg** — Bootstrap Bio “superbabies” investigation (2025-06-25). [bloomberg.com](https://www.bloomberg.com).
3. **STAT News** — NewLimit Series C (2026-06-02); Egli ethics commentary (2026-06-24); AbbVie-Calico split (2025-11-12).
4. **Science** — PGC vs Orchid data dispute [science.org](https://www.science.org); “Embryo editing first more complicated than headlines suggest” (2026).
5. **TechCrunch** — Nucleus Embryo backlash (2025-06-06); Retro \$1B raise (2025-01-24); NewLimit Series B (2025-05-06).
6. **Scientific American** — *Why genetically optimizing embryos is misleading, unethical — and not even possible* (2025).
7. **SCMP** — He Jiankui status reporting (2025).

Funding and industry databases

1. **Longevity.Technology investment reports** — 2025 total ~\$5.72B/170 deals; 2026 outlook. [longevity.technology](https://www.longevity.technology). *Note methodology-driven divergence from higher (\$8.5B) estimates.*
2. **agingbiotech.info** (Karl Pflieger) — 400+ company structured tracker.
3. **Company primary sources** (press releases; treat as self-reported): Gameto Series C [prnewswire.com](https://www.prnewswire.com); Conceivable Series A [globenewswire.com](https://www.globenewswire.com); Overture [businesswire.com](https://www.businesswire.com); Life Biosciences IND/dosing [lifebiosciences.com](https://www.lifebiosciences.com); NewLimit blog.newlimit.com; Altos CMO appointment altoslabs.com; XPRIZE Healthspan xprize.org; Immortal Dragons via lifespan.io; Hevolution disbursements via [Arab News](https://ArabNews.com).

Known verification gaps in this report

- Exact efficiency/mosaicism/off-target percentages for both June 2026 embryo-editing papers (paywalled primary texts).
- Current-year (FY2026) status of the US FDA heritable-modification appropriations rider.
- Retro Biosciences’ final closed amount; Conception’s total funding beyond ~\$20M; Herasight’s backers; an NCT identifier for RTR242.
- All company-reported birth/pregnancy tallies (Gameto, Conceivable) pending published trial data.

9. How to Keep This Current

Highest-signal X accounts (monitor daily/weekly): @antonioregalado (the single best source for embryo editing/IVG/PGS news breaking), @KarlPfleger (industry mapping), @EgliDieter and @NiakanLab (embryo-editing science), @ShaiCarmi and @SashaGusevPosts (statistical reality checks on embryo selection), @mkaeberlein (longevity skepticism), @jacobkimmel and @bettslacroix (reprogramming companies posting real data), @DinaRadenkovic/@GametoGen (Fertilo Phase 3), @pknoepfler (field watchdog), @sama and @bryan_johnson (capital and culture signals).

Journals (set alerts): *Nature*, *Nature Communications*, *Nature Medicine*, *Nature Aging*, *NEJM*, *Human Reproduction*, *Fertility and Sterility* (@FertStert), *Cell*, *GeroScience*, *npj Regenerative Medicine*.

Newsletters/outlets: MIT Technology Review's *The Checkup* (weekly), STAT News, Endpoints News (@endpts), Longevity.Technology weekly roundups, Lifespan.io's Rejuvenation Roundup, The Niche (ipsell.com).

Databases: agingbiotech.info, ClinicalTrials.gov saved searches ("partial reprogramming," "in vitro maturation," NCT07290244 and successors), HFEA and FDA advisory-committee calendars.

Events to watch for: Gameto Phase 3 readout, Life Bio ER-100 safety data, NewLimit IND, rentosertib Phase 3, the annual US appropriations rider fight, and any peer-reviewed publication from Conception or Paterna — the absence or arrival of those papers is itself the key signal.

Report compiled 2026-08-15 from sources dated through August 2026. Private-company figures are self-reported unless noted. This document is for research purposes and is not investment or medical advice.